Solution Requirements

Date	2
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Input	*Allow users to enter an event description and networking interests.*	High
2	Theme Extraction	*Extract key themes from the event description using the DistilBERT model.*	High
3	Conversation Starter Generation	*Generate personalized conversation starters using GPT-2 based on extracted themes and user interests.*	High

4	Fact Verification	*Verify user queries by retrieving reliable summaries from the Wikipedia API.*	High
5	Interaction History	*Save generated conversation starters and feedback history locally for future reference.*	Medium
6	User Interface	*Provide an interactive and responsive Streamlit interface for user interaction.*	High
7	Backend API	*Process requests using FastAPI and return responses efficiently.*	High
8	*Error Handling*	*Display meaningful error messages for invalid inputs or API failures.*	Medium

Step 2: Non-Functional Requirements (NFR)

S. No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**

1	Performance	*Generate conversation starters quickly after user input.*	*Response time less than 5 seconds (excluding first model loading).*
2	Scalability	*Support multiple user requests without affecting application performance.*	*Stable performance for concurrent users.*
3	Security	*Protect environment variables and validate user inputs.*	*Sensitive data stored in .env; input validation implemented*
4	Reliability	*Ensure consistent generation of conversation starters and fact verification.*	*Successful execution rate greater than 95%.*
5	Usability	*Provide a simple, intuitive, and responsive user interface.*	*Easy navigation with minimal user training.*
6	*Maintainability*	*Follow modular architecture with separate frontend, backend, AI services, and utilities.*	*Code organized into reusable modules with documentation.*